Bangladesh Health Data
ETL Project

Automated Health Data Pipeline using Excel & Power Query

29 Years
ANALYSIS PERIOD
(1994 - 2022)

20+
KEY INDICATORS
Tracked Over Time

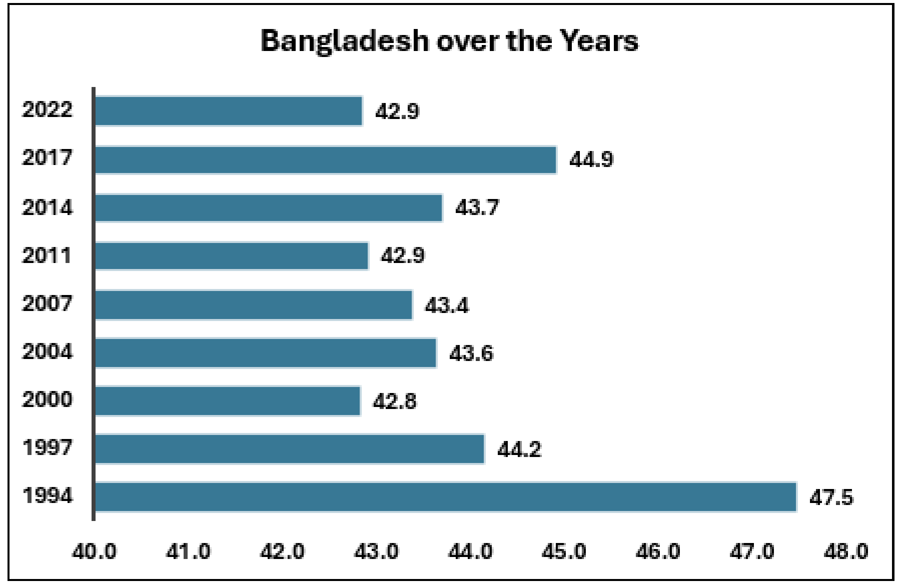
6 Themes
CATEGORIES
Standardized Taxonomy

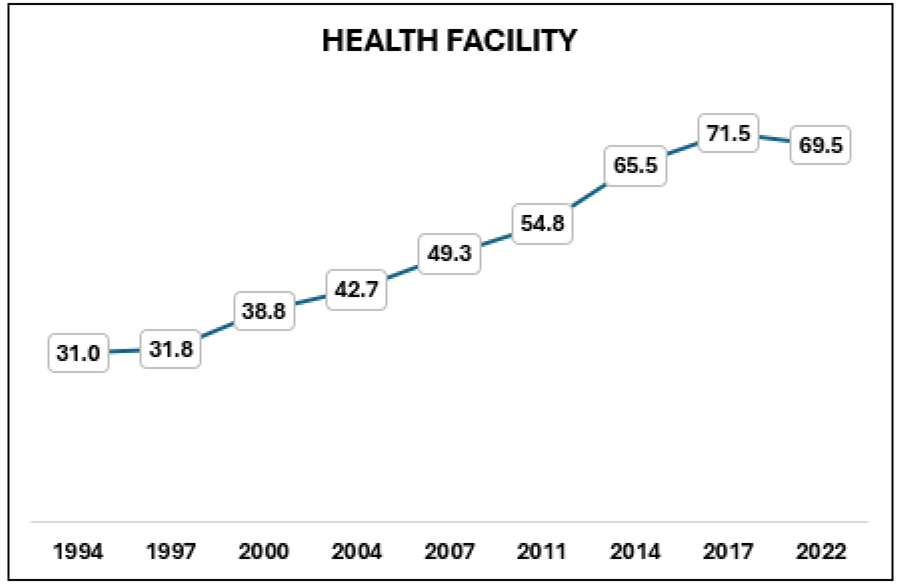
Primary Data Source: **The DHS Program (Health Surveys)**

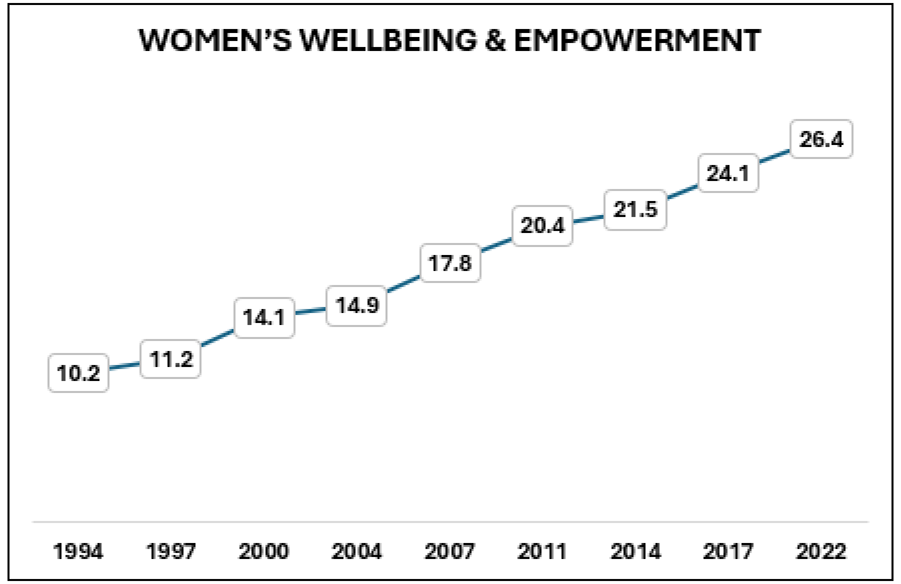
Trends in Reproductive, Maternal & Child Health of Bangladesh (1994–2022)

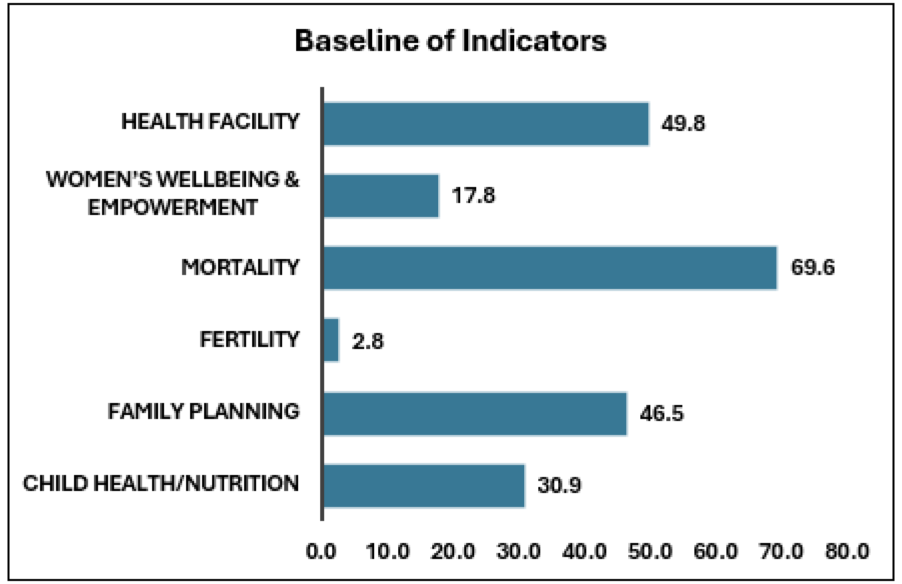
SurveyYear

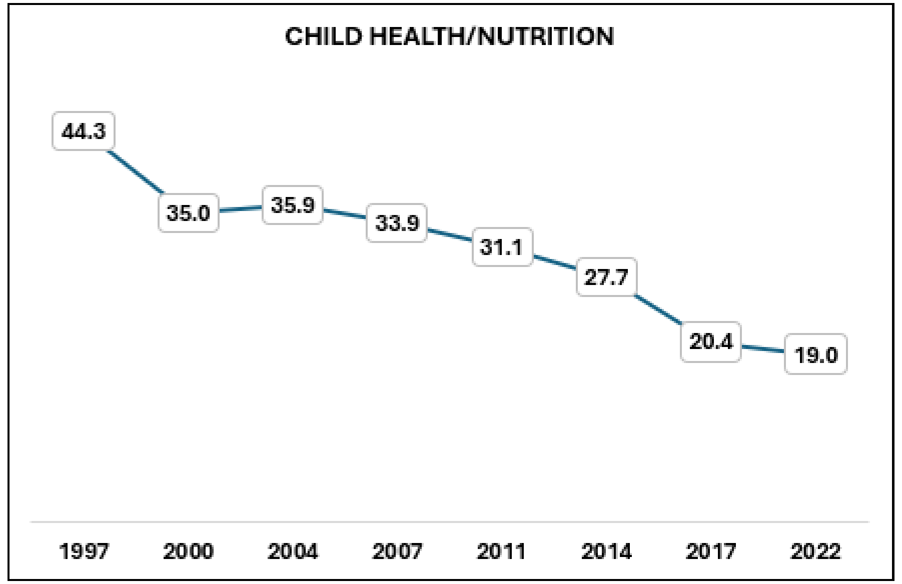
1994 | 1997 | 2000 | 2004 | 2007 | 2011 | 2014 | 2017 | 2022

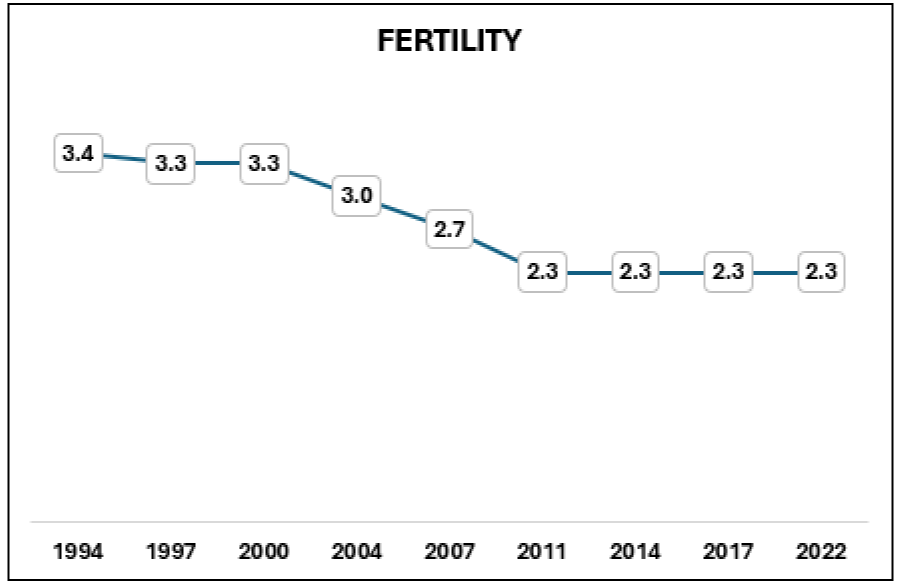


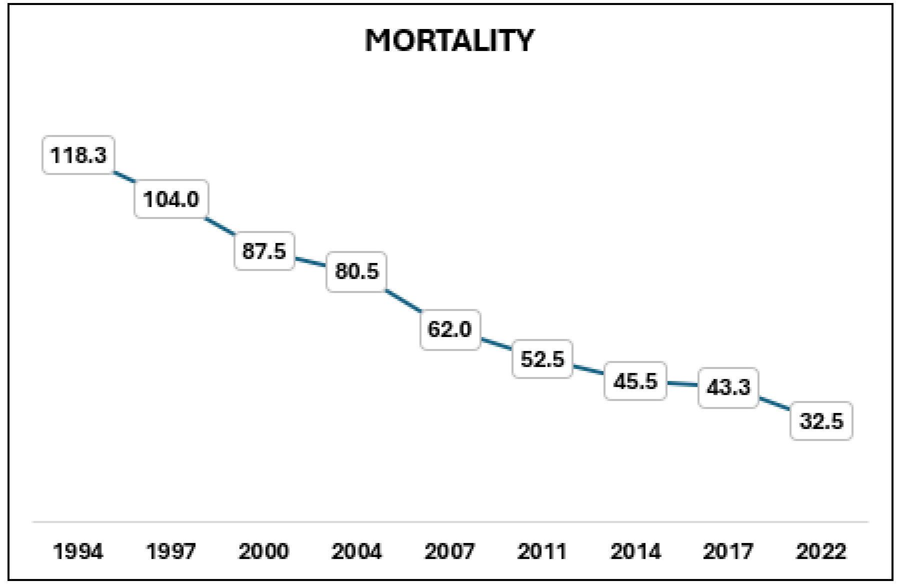


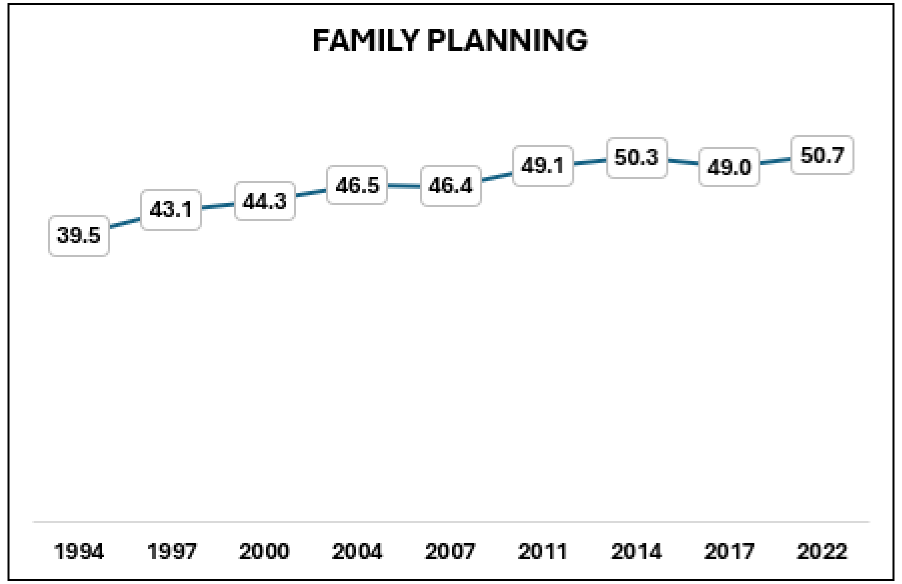












Project Overview

Challenge vs. Solution

The Challenge

Tracking long-term health trends in Bangladesh required manual data extraction from static files.

- Manual downloads
- Inconsistent data formatting
- Time-consuming update process
- Lack of interactive visualization

The Solution

An automated ETL pipeline using Excel Power Query that directly scrapes DHS QuickStats, transforms raw data into a tidy model, and powers a live-refresh dashboard.

Key Features

Web Scraping
Direct CSV URL extraction

ETL Automation
Repeatable transformation steps

Live Updates
One-click "Refresh All"

Categorization
6 standardized health themes

95%
TIME SAVED

29y
TREND CLARITY

Project Impact
Delivers consistent, error-free reporting with clear trend visuals for 1994–2022 without manual intervention.

ETL Pipeline Architecture

End-to-End Workflow

Extract
Data Ingestion

- **Source:** DHS Program QuickStats (Bangladesh)
- **Format:** Direct CSV URL access
- **Tool:** Excel Power Query (Get Data from Web)
- **Raw Volume:** 29 Columns, Mixed types

Transform
Data Processing

- **Reduction:** 29 → 4 core columns (Indicator, Value, Year, Category)
- **Cleaning:** Type coercion
- **Enrichment:** Map indicators to 6 themes (e.g., "TFR" → "Fertility")
- **Logic:** Rule-based text classification

Load
Visualization

- **Destination:** Excel Data Model Tables
- **Analysis:** Pivot Tables & Caches
- **Output:** Interactive Dashboard with Slicer
- **Refresh:** Live updates with single click

IMPLEMENTATION HIGHLIGHTS

One-Click Refresh | Web Scraping | Tidy Data Model

Data Categorization Strategy

6 Standardized Themes

Family Planning

- Contraceptive prevalence rate
- Unmet need for family planning
- Modern method usage

Child Health/Nutrition

- Stunting & Wasting prevalence
- Child mortality rates
- Vaccination coverage

Fertility

- Total Fertility Rate (TFR)
- Adolescent birth rate
- Birth spacing intervals

Mortality

- Infant mortality rate
- Under-5 mortality rate
- Neonatal mortality

Women’s Wellbeing

- Female education levels
- Household decision making
- Access to media

Health Facility

- Antenatal care (ANC) visits
- Institutional delivery rates
- Skilled birth attendance

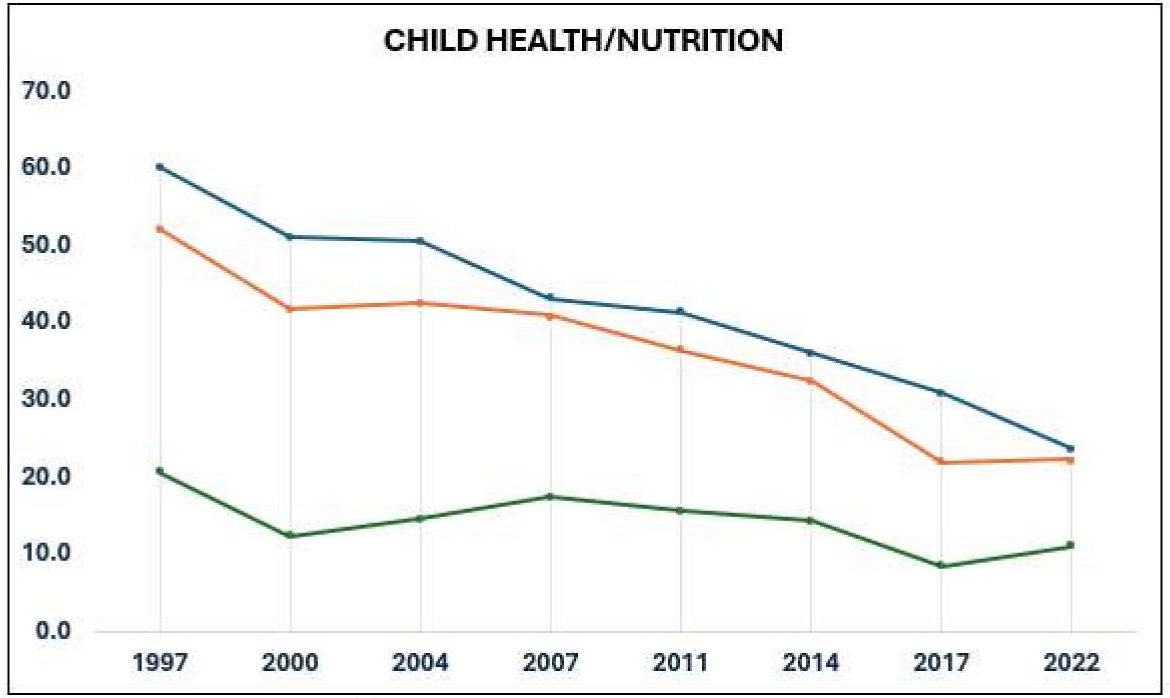
METHODOLOGY

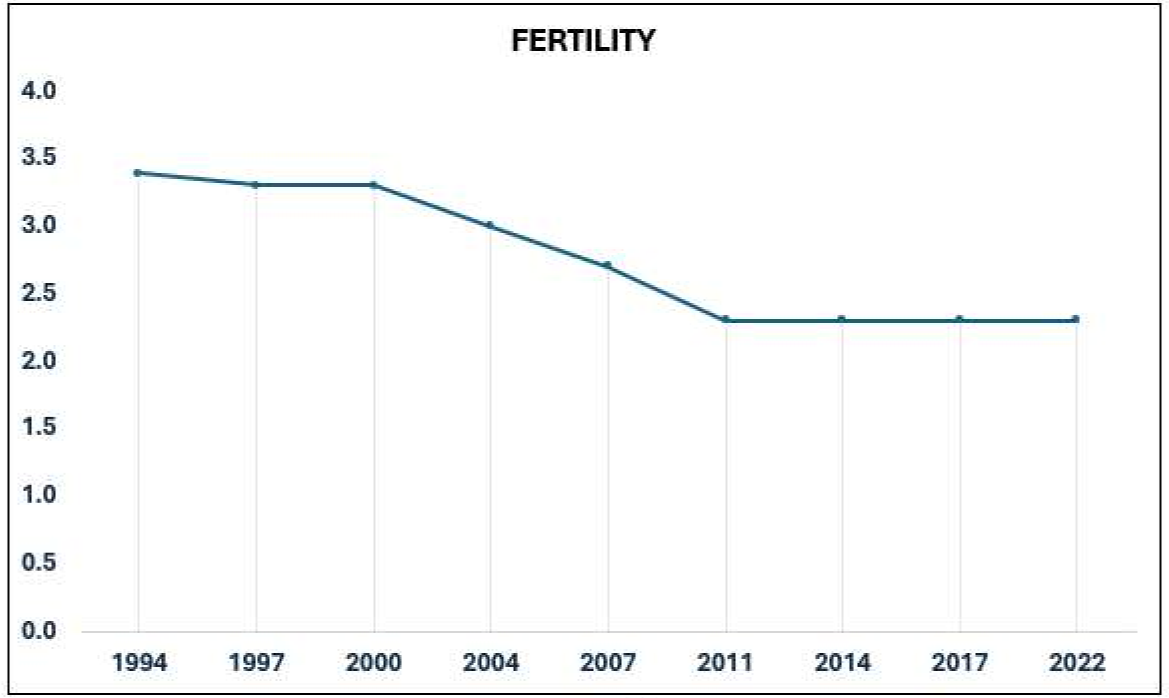
Rule-based mapping using keyword logic (Text.Contains) on indicator names to group 20+ raw metrics.

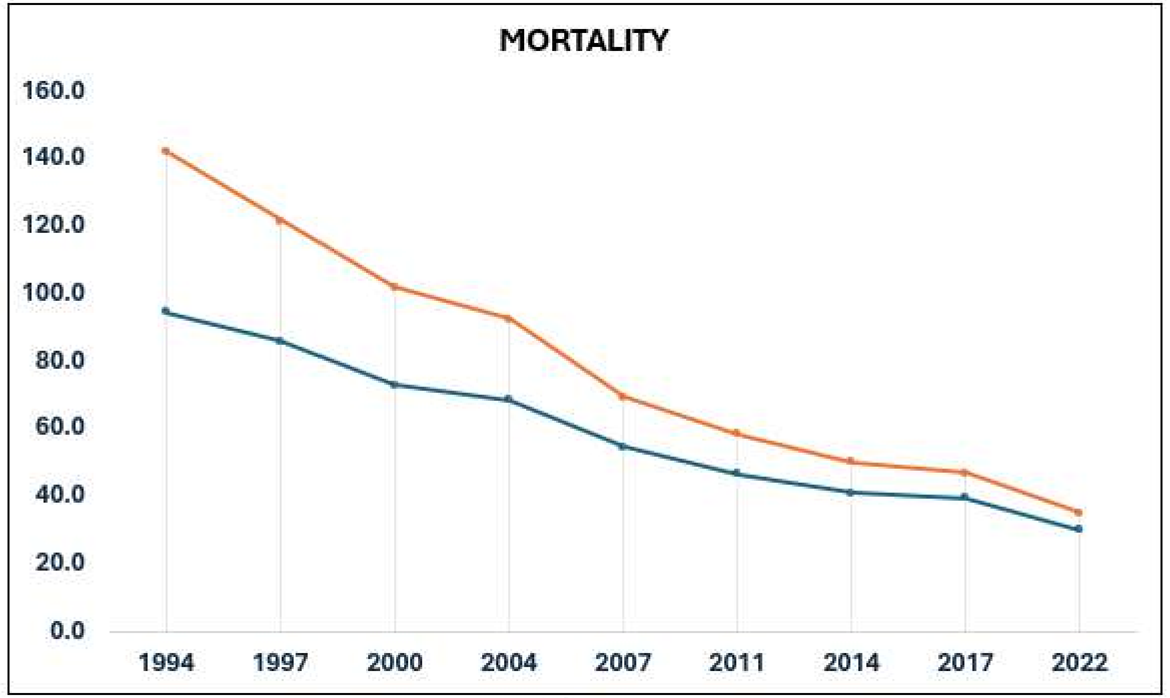
OUTCOME

Transforms unstructured data into a clean, thematic model ready for dimensional analysis and filtering.

Reproductive & Child Health Details





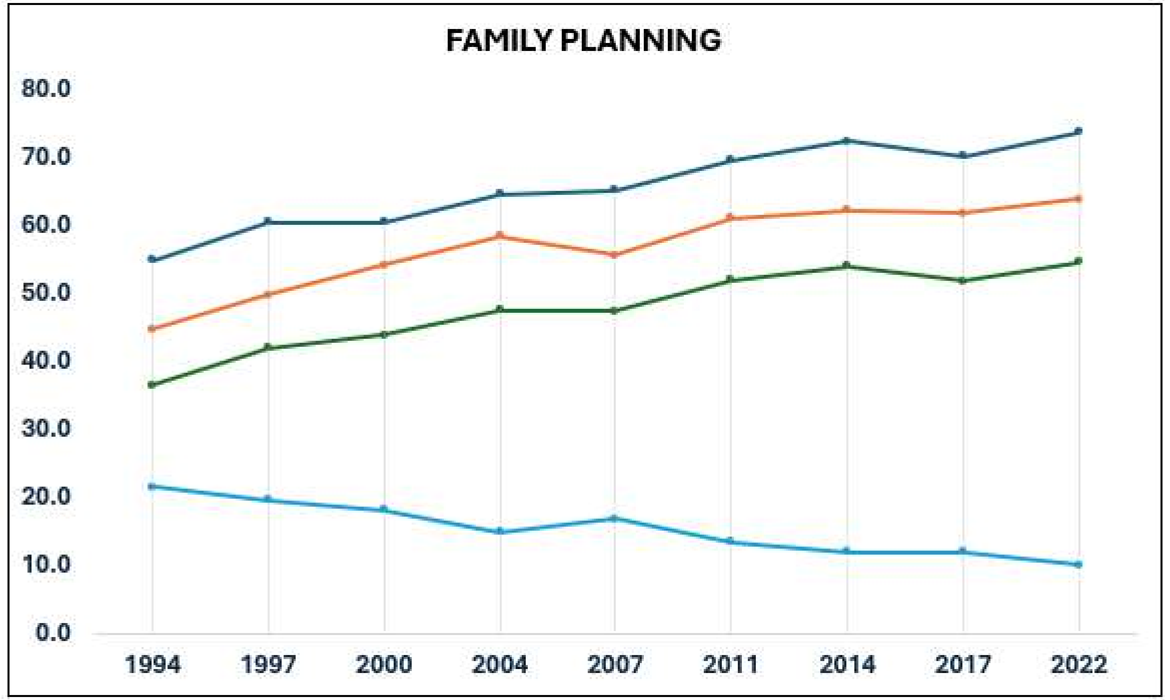


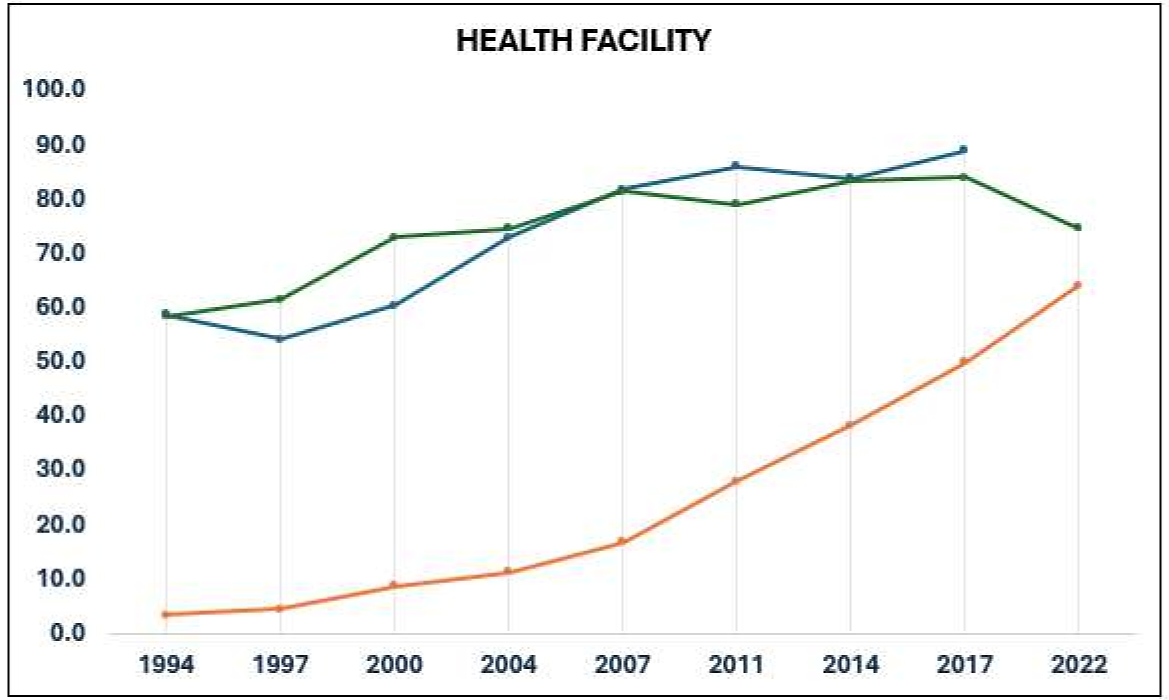
SurveyYe	Children stunted	Children underweight	Children wasted
1997	60.1	52.2	20.6
2000	51.1	41.7	12.3
2004	50.6	42.5	14.5
2007	43.2	41.0	17.4
2011	41.3	36.4	15.6
2014	36.1	32.6	14.3
2017	30.8	21.9	8.4
2022	23.6	22.3	11.0

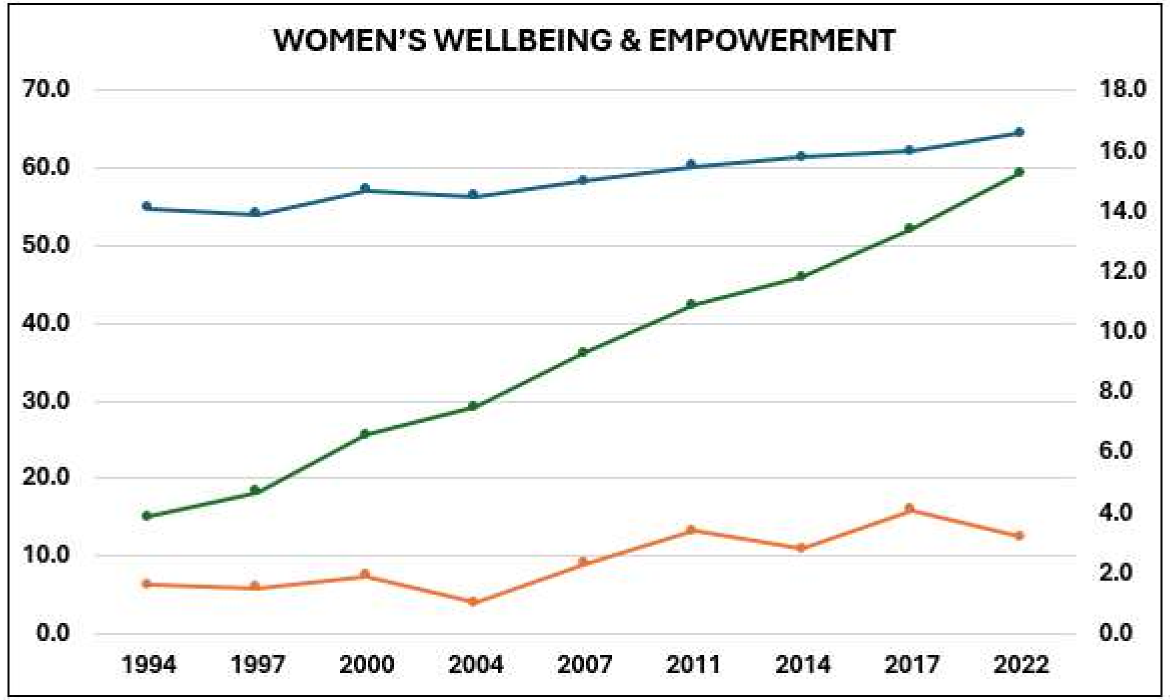
SurveyYea	Total fertility rate 15-49
1994	3.4
1997	3.3
2000	3.3
2004	3.0
2007	2.7
2011	2.3
2014	2.3
2017	2.3
2022	2.3

SurveyYe	Infant mortality rate	Under-five mortality rate
1994	94.5	142.0
1997	86.0	122.0
2000	73.0	102.0
2004	68.5	92.5
2007	54.5	69.5
2011	46.5	58.5
2014	41.0	50.0
2017	39.5	47.0
2022	30.0	35.0

Maternal Health Details







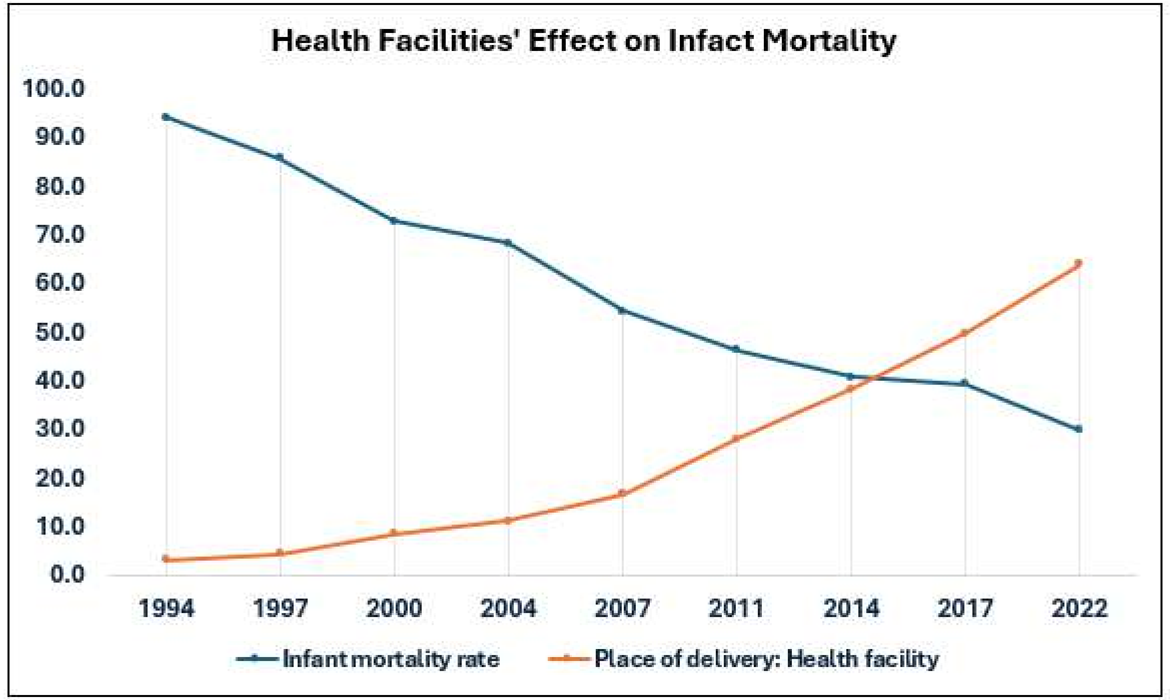
Survey Year	Demand for family planning satisfied by modern methods	Married women currently using any method of contraception	Married women currently using any modern method of contraception	Unmet need for family planning
1994	55.0	44.9	36.6	21.6
1997	60.6	49.8	42.1	19.7
2000	60.6	54.3	44.0	18.2
2004	64.8	58.5	47.6	15.0
2007	65.4	55.8	47.5	16.8
2011	69.7	61.2	52.1	13.5
2014	72.6	62.4	54.1	12.0
2017	70.3	61.9	51.9	12.0
2022	73.9	64.0	54.7	10.0

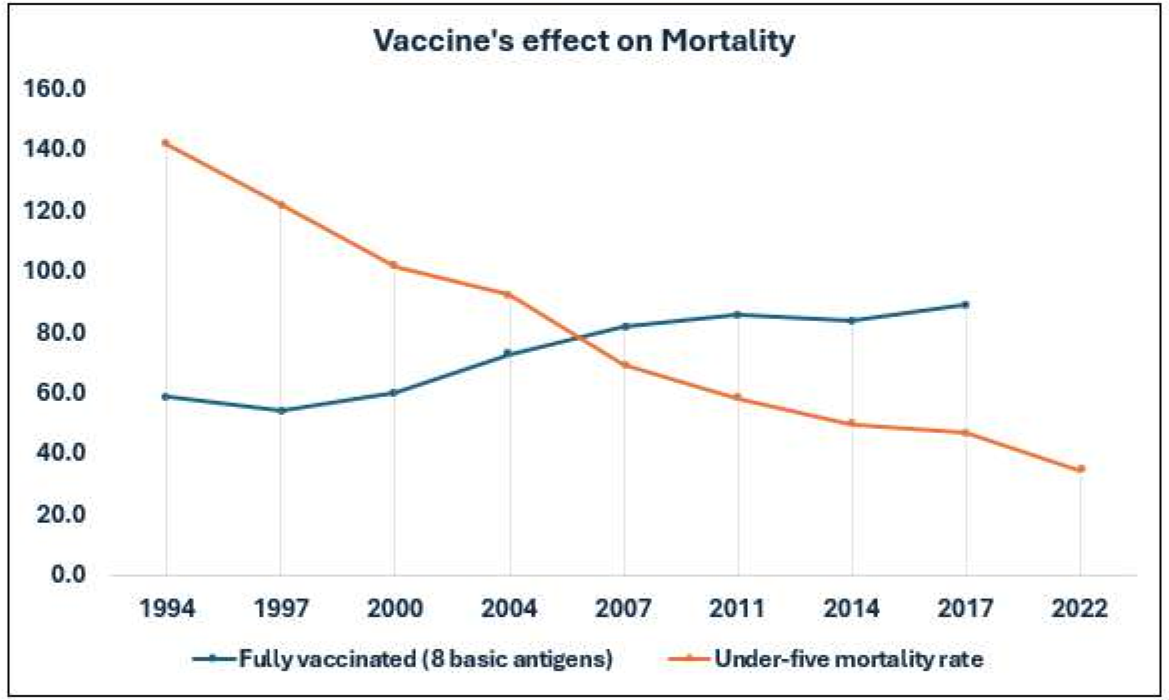
Survey Year	Fully vaccinated (8 basic antigens)	Place of delivery: Health facility	Treatment of diarrhea: Either ORS or RHF
1994	58.9	3.4	58.4
1997	54.2	4.6	61.6
2000	60.4	8.7	73.0
2004	73.1	11.3	74.7
2007	81.9	16.9	81.5
2011	86.0	28.1	79.1
2014	83.8	38.5	83.5
2017	89.1	50.0	84.2
2022		64.2	74.8

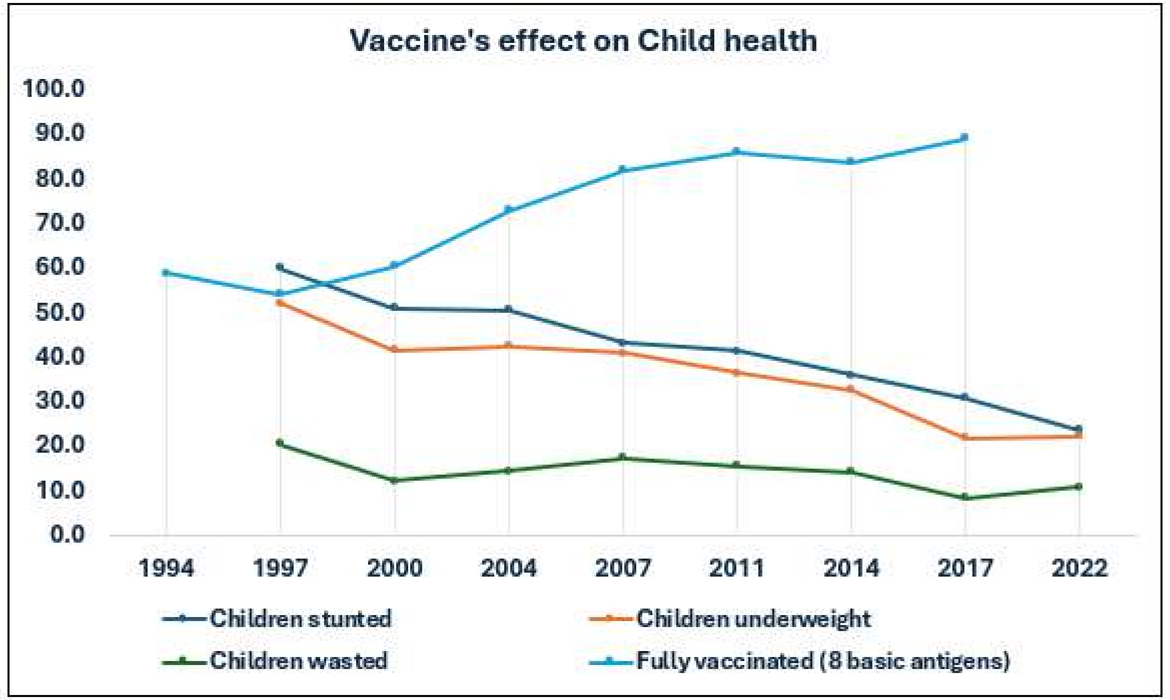
SurveyYear	Median age at first marriage [Women]: 25-49	Median duration of exclusive breastfeeding	Women with secondary or higher education
1994	14.1	1.6	15.0
1997	13.9	1.5	18.2
2000	14.7	1.9	25.6
2004	14.5	1.0	29.2
2007	15.0	2.3	36.2
2011	15.5	3.4	42.3
2014	15.8	2.8	45.9
2017	16.0	4.1	52.2
2022	16.6	3.2	59.4

Trends | Reproductive & Child Health | Maternal Health | Improvement of Child Health

Health Facilities' Effect on Child Health







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1994	58.9	142.0
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2017	89.1	47.0
2022		35.0

SurveyYear	Children stunted	Children underweight	Children wasted	Fully vaccinated (8 basic antigens)
1994				58.9
1997	60.1	52.2	20.6	54.2
2000	51.1	41.7	12.3	60.4
2004	50.6	42.5	14.5	73.1
2007	43.2	41.0	17.4	81.9
2011	41.3	36.4	15.6	86.0
2014	36.1	32.6	14.3	83.8
2017	30.8	21.9	8.4	89.1
2022	23.6	22.3	11.0	

Technical Skills Demonstrated

Core Competencies

Data Engineering

- Web scraping from trusted sources (DHS QuickStats)
- Automated ETL pipeline construction
- Building refreshable data connections without manual file handling

Power Query & M Language

- Connecting to web data: `Web.Contents`
- Parsing CSV structures: `Csv.Document`
- Advanced transformations: Type coercion, conditional columns, & text parsing

Data Modeling

- Dimensional modeling: Creating Category dimension
- Normalization: Reducing 29 raw columns to 4 tidy fields
- Optimizing data types for pivot table performance

Business Intelligence

- Dashboard UX/UI design & layout
- Creating interactive reports with Slicers & Pivot Charts
- Data storytelling & insight extraction

Additional Competencies

Data Quality QA | Reproducibility

PROJECT PURPOSE

To demonstrate end-to-end data capability from raw web source to executive dashboard.

Project Impact & Contact

Ready for Production

95%

TIME SAVED

Drastic reduction in reporting time vs. manual spreadsheet compilation.

Auto

LIVE UPDATES

Zero manual file handling; automated refresh from DHS API.

Scale

EXPANDABLE

Logic easily replicable for other countries or new health indicators.

100%

ACCURACY

Consistent definitions and automated type handling prevent errors.

Let's Connect

Interested in Excel ETL pipelines, or health analytics?

- LinkedIn
- GitHub
- Portfolio
- Email